[HW1 sols:]

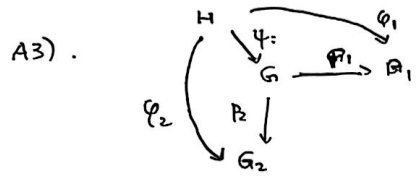
A)

A1) direct check

A2) direct check $\ker\pi_1 = \{(1, g_2) \mid g_2 \in G_2\}$

$\ker\pi_2 = \{(g_1, 1) \mid g_1 \in G_1\}$.

A3).



$\psi: H \longrightarrow G_1 \times G_2$

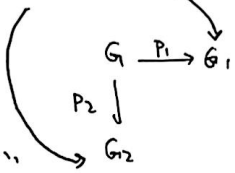
$h \longmapsto (\varphi_1(h), \varphi_2(h))$ $\begin{cases} \varphi_1(h) \in G_1, \\ \varphi_2(h) \in G_2. \end{cases}$

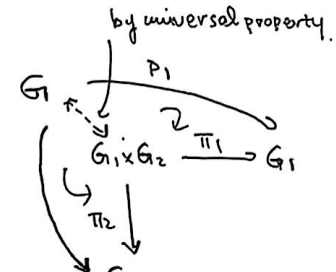
pick $G = G_1 \times G_2$. $p_i: G_1 \times G_2 \longrightarrow G_i$ (π_i in A2).

$(g_1, g_2) \longmapsto g_i$

Check: uniqueness of ψ.

check: uniqueness of G





check $G \simeq G_1 \times G_2$ and commutativity

A4). $\phi: \mathbb{Z}/_{n_1 n_2 \mathbb{Z}} \longrightarrow \mathbb{Z}/_{n_1\mathbb{Z}} \times \mathbb{Z}/_{n_2\mathbb{Z}}$

$\overline{k} \longmapsto (k \bmod n_1,\ k \bmod n_2)$

check: "morphism" "injective" "surjective"

A5) $\mathbb{Z}_p \times \mathbb{Z}_p$. no order p^2 element

$\Rightarrow \mathbb{Z}_p \times \mathbb{Z}_p \not\simeq \mathbb{Z}_{p^2}$ so ~~does~~ isn't cyclic

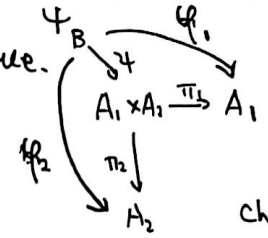
A6). direct check: • π_i: ring hom

• $A_1 \times A_2$ ring (check def of ring).

A7). • uniqueness: A4). additive group $(A, +)$

uniqueness of

• existence.



$\psi: B \longrightarrow A_1 \times A_2$

$b \longmapsto (\varphi_1(b), \varphi_2(b))$. so $\exists$ such ψ

Check: • ring hom

• well-defined.

A8) $\mathbb{Z}/_{mn\mathbb{Z}} \longrightarrow \mathbb{Z}/_{m\mathbb{Z}} \times \mathbb{Z}/_{n\mathbb{Z}}$

$\overline{k} \longmapsto (\overline{k} \bmod m,\ k \bmod n)$

check: ring hom

(+, ×)

ring iso (injective + surjective)

↑

CRT ← Chinese remainder theorem

A9).

$\psi: (A \times_{\text{ring}} B)^\times \simeq A^\times \times_{\text{group}} B^\times$

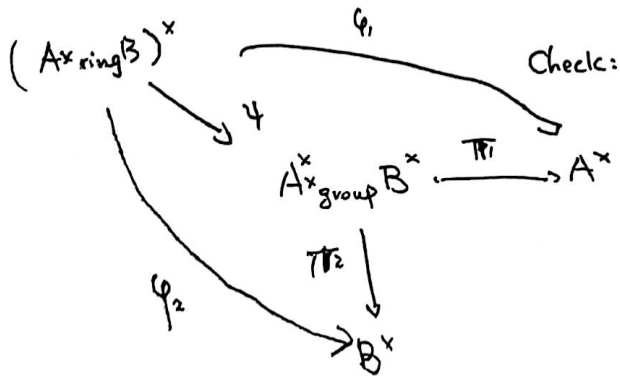
$(a, b) \longmapsto (a, b)$

$\varphi_1: (A \times_{\text{ring}} B)^\times \longrightarrow A^\times$

$(a, b) \longmapsto a$

$\varphi_2: (A \times_{\text{ring}} B)^\times \longrightarrow B^\times$

$(a, b) \longmapsto b$



Check: • φ_1, φ_2 • well-defined • group hom

} universal property $\Rightarrow \psi$:

Check: ψ inj + surj

$\Rightarrow \psi$ iso $\Rightarrow$

$(A \times_{\text{ring}} B)^\times \simeq (A^\times \times_{\text{group}} B^\times)$

B)

B1) $\mathbb{Z}/n\mathbb{Z}$ $\quad k\in(\mathbb{Z}/n\mathbb{Z})^\times \Rightarrow \exists\, l\ \ kl\equiv 1 \pmod n$

$\Rightarrow (k,n)=1.$

if $(k,n)=1 \overset{\text{Bezout}}{\Rightarrow} \exists\, l,s\ \ kl+ns=1 \Rightarrow k\in(\mathbb{Z}/n\mathbb{Z})^\times \Rightarrow |\mathbb{Z}/n\mathbb{Z}^\times| = \phi(n)$

B2) A9 $\quad |(A\times B)^\times| \overset{A9}{=} |A^\times \times B^\times| = |A^\times||B^\times| = \phi(m)\phi(m)$

(A: ring, B: field; $\times$: group)

$\phi(mn)$.

$\mathbb{Z}/n\mathbb{Z}^\times \xleftrightarrow{1:1} \{k \mid (k,n)=1\}$

$\phi(n) = \phi(\Pi_i p_i^{a_i}) = \Pi_i \phi(p_i^{a_i}) = \Pi_i p_i^{a_i}\cdot\frac{p_i-1}{p_i} = n\prod_i(1-\frac{1}{p_i})$

B3) $a\in(\mathbb{Z}/n\mathbb{Z})^\times$ since $(a,n)=1$, $|\mathbb{Z}/n\mathbb{Z}^\times| = \phi(n)$ $\Rightarrow$ $a^{\phi(n)} \equiv 1 \pmod n$ (when $n=p \overset{\phi}{\Rightarrow} a^{p-1}\equiv 1 \pmod p$ (Fermat)).

B4) $\mathbb{Z}/n\mathbb{Z}$ generator g $\quad C_d = \langle g^{\frac{n}{d}} \rangle$ $\leftarrow$ $\frac{n}{d}\in\mathbb{Z}$ well-defined.

- Check: order $(g)=d$ it is actually C_d.
- uniqueness. $\nexists\, \mathbb{Z}/n\mathbb{Z} \rightsquigarrow |\mathbb{Z}/d\mathbb{Z}| = d$ so

l and since if an element in $C_d \Rightarrow l^d = 1$

$\Rightarrow l^d = g^{kn}$ for some $k\in\mathbb{Z}$

$\Rightarrow l = g^{k\frac{n}{d}} \Rightarrow l = \left(g^{\frac{n}{d}}\right)^k$

5) $n=\Pi_i p_i^{a_i}$ $\Rightarrow$ if $l\in C_d \Rightarrow l\in\mathbb{Z}\langle g^{\frac{n}{d}}\rangle$ so only 1 C_d.

$\phi(n) = n\Pi_i(1-\frac{1}{p_i})$.

only do for p^a case. $\phi(p^a) = p^{a-1}(p-1)$

$\Downarrow$

$1+\phi(p^1)+\cdots+\phi(p^a) = (1+\cdots+p^{a-1})(p-1)+1 = p^a$ ✓

using $\phi(m)\phi(n)=\phi(mn)$ to generalize to all n

6)

$n = \sum_{d\mid n, G_d\neq\phi} \phi(d)$ $\quad G<k^\times$ $|G|<\infty$ $|G|=n$.

Aim: ~~$d\mid n \Leftrightarrow G_d\neq\phi$~~

pick $g\in G$. $\mathrm{ord}(g)=d$

$\Rightarrow g^d = e$ has $\phi(d)$ elements s.t. order $= d$. ($d, l\neq 1$ for g^l)

$\Rightarrow \langle g\rangle$

~~uniqueness~~

$\Rightarrow$ if order d elements exists, then has $\phi(d)$ (at least).

$\Rightarrow n \le \sum_{d\mid n, G_d\neq\phi} \phi(d)$.

每个元素都会被算在内

- $n = \sum_{d\mid n}\phi(d) \geq \sum_{d\mid n, G_d\neq\phi}\phi(d)$ by def

$\Rightarrow$ "="

B7) $G_n\neq\phi \Rightarrow n$ cyclic

B8) view $\mathbb{Z}/p\mathbb{Z}$ as $\mathbb{F}_p$ $\quad \mathbb{F}_p^\times$ is cyclic

B9) $\left(\mathbb{Z}/p^m\mathbb{Z}\right)^\times \quad -(1+p)^{p^{k-2}} \equiv 1+p^{k-1} \pmod{p^k}$

induction $1+p$

$k=0$ ✓ $\quad k\le n-1$ ✓ $\quad (1+p)^{p^n} \equiv 1+p^{n+1} \pmod{p^{n+2}}$

$\Rightarrow (1+p^{p^n})^{p^n} \pmod{p^{n+2}}$

$(1+p)^{p^n} \equiv$

s

$(1+p^n+sp^{n+1})^p$ $\quad s\in\mathbb{Z}$

$=$

$1+p(p^n+sp^{n+1}) + C_p^2(p^n+sp^{n+1})^2 + \cdots$

cancel under $\equiv p^{n+2}$

$\Downarrow$

$\equiv 1+p^{n+1} \pmod{p^{n+2}}$

- $\mathbb{Z}/p^m\mathbb{Z}$ $(1+p)^{p^{m-1}} \equiv 1+p^m \pmod{p^{m+1}}$.

$\Rightarrow (1+p)^{p^{m-1}} \equiv 1 \pmod{p^m}$ $\Rightarrow$ order$(1+p) \mid p^{m-1}$ if not p^{m-1}

$(1+p)^{p^{m-2}} \equiv 1+p^{m-1} \pmod{p^m}$ so $\text{ord}(1+p) = p^{m-1}$.

- $(\mathbb{Z}/p\mathbb{Z})^\times$ generator a. then if $a^l \equiv 1 \pmod p \Rightarrow p-1 \mid l$

$a^l \equiv 1 \pmod{p^m} \Rightarrow p-1 \mid l$. suppose $l' = \text{ord}(a)$ consider $a^{\frac{\text{ord}(a)}{p-1}}$ ✓

put in $\in (\mathbb{Z}/p^m\mathbb{Z})^\times$

- $\bar{l} = \overline{k(p+1)}$

$\text{ord}(\bar{l}) = \text{lcm}(\bar{k}, \overline{p+1}) = \text{lcm}(p-1, p^{m-1}) = (p-1)p^{m-1}$.

$|\mathbb{Z}/p^m\mathbb{Z}| = \Phi(p^m) = p^{m-1}(p-1)$ $\Rightarrow \langle \bar{l} \rangle = (\mathbb{Z}/p^m\mathbb{Z})^\times$

B10). $(\mathbb{Z}/2^m\mathbb{Z})^\times$

$(1+2^2)^{2^k} \equiv 1+2^{k+2} \pmod{2^{k+3}}$

$k=1$ $25 \equiv 9 \pmod{16}$

$k=0$ $5 \equiv 5 \pmod 8$ ✓

- $(1+2^2)^{2^{l+1}} = \left(1+2^{l+2}+l_2 2^{l+3}\right)^2 = 1+2\left(2^{l+2}+l_2 2^{l+3}\right)+\left(2^{l+2}+l_2 2^{l+3}\right)^2$

$\equiv 1+2^{l+3} \pmod{2^{k+4}}$

$\Rightarrow$ induction ✓

$\bar{5} = (1+2^2)^{2^{m-2}} \equiv 1+2^m \pmod{2^{m+1}}$ so is $\bar{1}$ in $\mathbb{Z}/2^m\mathbb{Z}$

$(1+2^2)^{2^{m-3}} \equiv 1+2^{m-1} \pmod{2^m}$ $\Rightarrow \text{ord}(5) = 2^{m-2}$

- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{m-2}\mathbb{Z} \longrightarrow (\mathbb{Z}/2^m\mathbb{Z})^\times$

$(a,b) \longmapsto (-1)^a 5^b$.

ker is $(-1) \notin \langle \bar{5} \rangle$

- check: • well-defined (need above)

• group homo

• $|\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{m-2}\mathbb{Z}| = |(\mathbb{Z}/2^m\mathbb{Z})^\times| = 2^{m-1}$.

• $a-1 \notin \langle \bar{5} \rangle$ if yes $-1 \in \langle \bar{5} \rangle$.

$\text{ord}(5) = 2^{m-2}$. and $(-1)^2 = 1 \Rightarrow -1 = 5^{2^{m-3}}$ in $\mathbb{Z}/2^m\mathbb{Z}$

$-1 \equiv 5^{2^{m-3}} \equiv 1+2^{m-1} \pmod{2^m}$

by 1 not -1 so gives contradiction.

• injective ✓

11). $(\mathbb{Z}/n\mathbb{Z})^\times = \prod_i \left(\mathbb{Z}/p_i^{a_i}\mathbb{Z}\right)^\times$ ← 互為一个非平凡

$n = \prod_i p_i^{a_i}$

trivial group given by. $(\mathbb{Z}/2\mathbb{Z})^\times$

So $n = 2p^\alpha$

- p^α B9). $(\mathbb{Z}/p^m\mathbb{Z})^\times$ cyclic
- 2^β B10) $(\mathbb{Z}/2^m\mathbb{Z})^\times$ cyclic iff $\forall\, m \le 2$. So $\boxed{n = 1, 2, 4, p^m, 2p^m,}$

def of "原根" := generator of $(\mathbb{Z}/n\mathbb{Z})^\times$

C. $M_n(\mathbb{Z})$.

$A A^* = \det(A) \cdot I$. so if $\det(A) = \pm 1 \Rightarrow A^* = \pm A^{-1} \in M_n(\mathbb{Z})$.

- $GL_n(\mathbb{Z}) = \{A \in M_n(\mathbb{Z}) \mid \det(A) = \pm 1\}$ otherwise $A^{-1} = \frac{\pm 1}{\det A} A^*$ not in $M_n(\mathbb{Z})$.

↑ view in $\mathbb{Q}$

- If $A \in GL(2;\mathbb{Z})$ key point: $A \in GL(n)$ and. $A^m = I \Rightarrow$ A |eigv of A| $= 1$

A has form $P \begin{pmatrix} \mu_1 & \\ & \mu_2 \end{pmatrix} P^{-1}$ ← unit root. A diagonalizable

consider $A \in GL(2;\mathbb{Z})$ A satisfies $A^2 + kA + \bar{5} I_2 = 0$ ← Cayley Hamilton

• in $GL(2;\mathbb{Z})$

$\Rightarrow \det = 1 \Rightarrow s = \pm 1 \Rightarrow$ suppose $s=1$, A satisfies $A^2 + kA + I = 0$ ← pick $k \in \mathbb{Z}$, check.

eigenvalue are root of $x^2 + kx + 1 = 0$.

$k = -2 \sim 2 \Rightarrow$ gives order 1, 2, 3, 4, 6

So λ must be unit root with cyclotomic polynomial deg $\lambda^m = 1$ + $\varphi(m) \le 2 \Rightarrow m = 1, 2, 3, 4, 6$

~~(If $\det A = -1$, then $2 \mid \mathrm{ord}(A)$ and consider $-A$)~~

if $\det A = -1$ ~~$\Rightarrow \det$~~

$\Rightarrow 2 \mid \mathrm{ord}(A)$ consider changing sign of one eigenvalue

• $\forall A \in GL(n;\mathbb{Z})$ $\exists C_n$

$\mathrm{ord}(A) \le C_n$. eig of A is primitive unit root. $\Rightarrow$ consider deg $= m$ (called λ)

$\uparrow\uparrow$

$\mathrm{ord}(A) < \infty$

λ has $\Phi_m(X)$ as minimal polynomial in $\mathbb{Q}(x)$

↑ cyclotomic polynomial.

• $\deg \Phi_m(X) = \varphi(m)$

• $\deg \Phi_m(X) \le n$. ⇝ by Cayley Hamiltonian $P_A(\lambda) = 0$ ← Char poly of A

⇓

and C_n = product of m s.t. $\varphi(m) \le n$ i.e. $C_n = \prod_{\varphi(m) \le n} m$ only finitely many m $\Rightarrow \boxed{\Phi_m(X) \mid P_A(X)}$ ☆

$\Rightarrow \lambda^{C_n} = 1 \Rightarrow A^{C_n} = I_n$.

): (Cancellation law)

3-thm: refer to any Abstract algebra book.

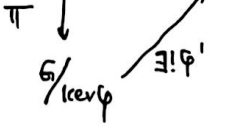
$U(G,G') = |\mathrm{Hom}(G,G')|$ $I(G,G') = |\{\varphi \in \mathrm{Hom}(G,G') \mid \varphi \text{ injective}\}|$

$\mathrm{Hom}(G,G')$ = group hom of G to G'

D1)

$$M(G,G') = \sum_{H \lhd G} I(G/H, G')$$

$\varphi \in \mathrm{Hom}(G,G')$



$\varphi \rightsquigarrow \varphi'$ for $H = \ker\varphi$

so LHS ≤ RHS

$\forall \varphi \in \mathrm{Hom}(G/H, G')$ then $\overline{\varphi} = \varphi \circ \pi$ gives $G \to G'$ map

check: all these maps are different ⇒ RHS ≥ LHS

then $M(G,G') = \sum_{H \lhd G} I(G/H, G')$

D2) $I(G,G') = \sum_{H \lhd G} \lambda_H \cdot M(G/H, G')$ ↑ indep of G' $= \sum_{H \lhd K \lhd G} \lambda_H \cdot I(G/K, G')$.

• $M(G/H, G') = \sum_{H \lhd K \lhd G} I(G/K, G')$

C2) p prime. $q = p^n$. $\mathbb{F}_q$

Cayley-Hamilton ⇒ $P_A(A) = 0$ $\deg P_A = n$

$- \{P(A) \mid P \in \mathbb{F}_q[x]\}$

$= \{R(A) \mid R \in \mathbb{F}_q[x] \deg R \le n-1\}$ $(A \in GL(n;\mathbb{F}_q))$.

$P(x) = P_A(x)Q(x) + R($

$\Rightarrow |\{P(A) \mid P \in \mathbb{F}_q[x]\}| \le |\mathbb{F}_q^n|$ $\Rightarrow P(A) = R(A)$

$- \langle A \rangle \subseteq \{P(A) \mid P \in \mathbb{F}_q[x]\}$ $\overset{\text{finite}}{\subseteq} q^n$ $|\langle A \rangle| \le q^n - 1$

$- L = \mathbb{F}_{q^n} / \mathbb{F}_q = k$ then consider. $L \to L$, $x \to yx$ $y \in L$

$L^\times$ cyclic group pick y generator

So such example exists.

A $I(G,G')$ (G/H normal subgrp has form K/H $(G/H)/(K/H) \simeq G/K$. ↑ 3-rd isom thm)

$M(G/H, G') = \sum_{H \lhd K \lhd G} I(G/K, G')$.

So it suffices to show $I(G,G') = \sum_{H \lhd K \lhd G} \lambda_H I(G/K, G')$.

let $(S_{ij})_{ij} = \begin{cases} 1 & H_i \lhd H_j \\ 0 & \text{otherwise} \end{cases}$

$\Rightarrow M(G/H_i, G') = \sum_{j=1}^{n} S_{ij} I(G/H_j, G')$.

by setting $H_1, \dots, H_n$ appropriately can s.t. it is upper-triangular + on diagonal all elements are 1 (s.t. $\forall j > i$ $H_j \not\supset H_i$).

$S_{ij} = \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix}$ ↑ n = number of normal subgrp.

pick $\lambda_1, \dots \lambda_n$ s.t. $\lambda_1 d_1 + \dots + \lambda_n d_n = e_1$

$\lambda_H := \lambda_K$ if H is set to H_K in rearrangement.

exist + unique.

$\Rightarrow$ direct check gives $I(G,G') = \sum_{H \lhd K \lhd G} \lambda_H I(G/K, G')$ ✓

$\{e\}$ = H_1 H_2 $\cdots$ H_K $\cdots$ H_n = G_1

D3) $I(G_1 \times G', G_1') \simeq I(G_1', G_2)$.

改为

$\Updownarrow$

$\sum_{H \lhd G} \lambda_H M(G'/H, G_1)$

$\Updownarrow$

$\sum_{H \lhd G} \lambda_H M(G'/H, G_2)$

$\Updownarrow$

$I(G', G_2)$

$|\mathrm{Hom}(G'/H, G_1 \times G')| = |\mathrm{Hom}(G'/H, G_2 \times G')|$

using (A) $\Downarrow$ since $G_1 \times G' \simeq G_2 \times G'$

$|\mathrm{Hom}(G'/H, G_1)||\mathrm{Hom}(G'/H, G')| = |\mathrm{Hom}(G'/H, G_2)| \cdot |\mathrm{Hom}(G'/H, G')|$

$\Downarrow$

$|\mathrm{Hom}(G'/H, G_1)| = |\mathrm{Hom}(G'/H, G_2)|$ $H \lhd G$

1) $G_1 \times G' \simeq G_2 \times G'$

$\Rightarrow |\mathrm{Hom}(K, G_1)| = |\mathrm{Hom}(K, G_2)|$ $\forall K$. Since $|\mathrm{Hom}(K, G_1 \times G')| = |\mathrm{Hom}(K, G_2 \times G')|$

$\Rightarrow M(K, G_1) = M(K, G_2)$ $\forall K$

$\Rightarrow I(K, G_1) = I(K, G_2)$ $\forall K$ using D2)

$\Rightarrow I(G_1, G_2) \circ I(G_2, G_1) \geq 1$ $\Rightarrow$ $\exists$ injective $G_1 \to G_2$, injective $G_2 \to G_1$ $\Rightarrow |G_1| = |G_2|$ and $G_1 \hookrightarrow G_2$

≥ 1 = id_{G_1} $I(G_1, G_1)$; $I(G_2, G_2)$ id_{G_2}

$\Rightarrow G_1 \simeq G_2$

) $[G_1 : \mathbb{F}_2] = m$ $[G_2 : \mathbb{F}_2] = n$. G_1, G_2 view as dim m, n $\mathbb{F}_2$-V.S.

$\varphi: G_1 \times G' \to G_2 \times G'$ • $m > n$.

• $(a_1 \dots a_m) \times (b_0 \dots) \longmapsto (a_1 \dots a_n) + (a_{n+1}, a_{n+2} \dots a_m, b_0 \dots)$.

• φ keeps "+" $\ker \varphi = \{0\}$

• $G_1 \not\simeq G_2$.

so need finite.